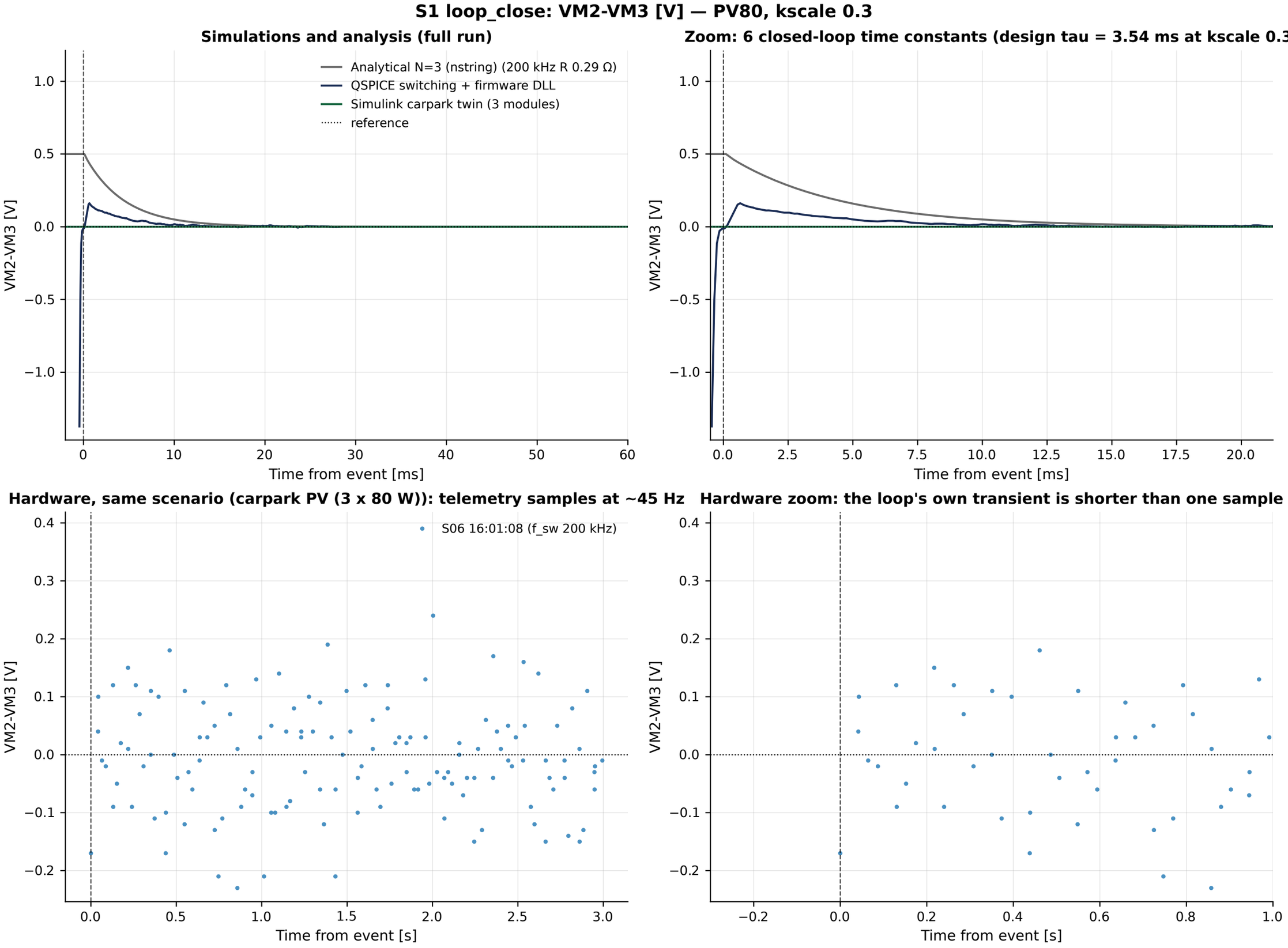
S1 loop_close: VM2-VM3 [V] — PV80, kscale 0.3
Simulations and analysis (full run)
Analytical N=3 (nstring) (200 kHz R 0.29 Ω)
QSPICE switching + firmware DLL
Simulink carpark twin (3 modules)
reference
VM2-VM3 [V]
Time from event [ms]
Zoom: 6 closed-loop time constants (design tau = 3.54 ms at kscale 0.3
VM2-VM3 [V]
Time from event [ms]
Hardware, same scenario (carpark PV (3 x 80 W)): telemetry samples at ~45 Hz
S06 16:01:08 (f_sw 200 kHz)
VM2-VM3 [V]
Time from event [s]
Hardware zoom: the loop's own transient is shorter than one sample
VM2-VM3 [V]
Time from event [s]